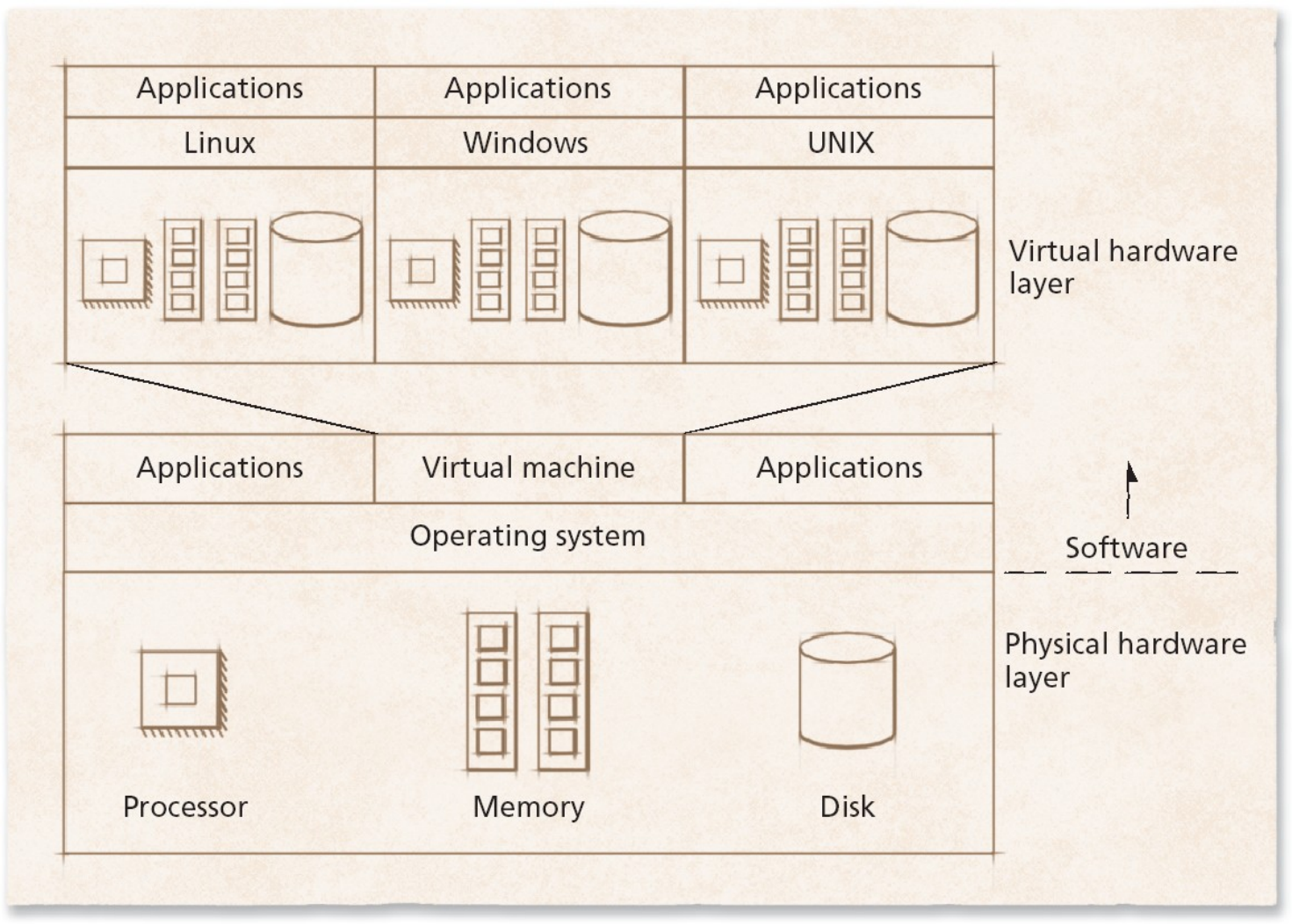
Operating Systems

Operating System Environments

- Virtual machines (VMs)
  - Software abstraction of a computer
  - Often executes on top of native operating system
- Virtual machine operating system
  - Manages resources provided by virtual machine
- Applications of virtual machines
  - Allow multiple instances of an operating system to execute concurrently
  - Emulation
    - Software or hardware mimics functionality of hardware or software not present in system
  - Promote portability

Operating System Environments

Figure 2 Schematic of a virtual machine.



Operating System Architectures

- Today’s operating systems tend to be complex
  - Provide many services
  - Support variety of hardware and software
  - Operating system architectures help manage this complexity
    - Organize operating system components

KERNAL

- Kernel is the core part of an operating system; it manages the system resources. Kernel is like a bridge between application and hardware of the computer. The Kernel can be classified further into two categories, Microkernel and Monolithic Kernel. Microkernel is the one in which user services and kernel services are kept in separate address space. However, in Monolithic kernel user services and kernel services both are kept in the same address space

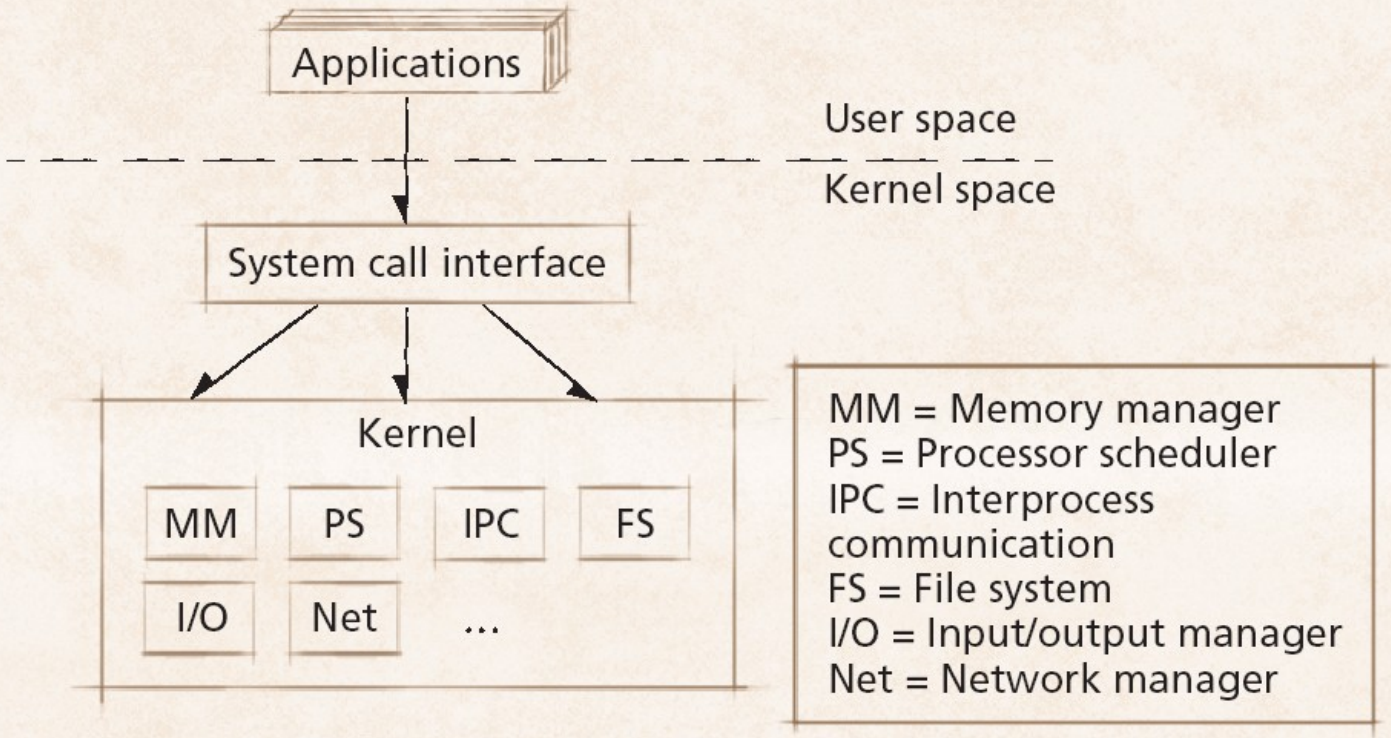
Monolithic Architecture

- Monolithic operating system
  - Every component contained in kernel
    - Any component can directly communicate with any other
  - Tend to be highly efficient

- The entire O.S. is placed inside the kernel
- It runs as a single large process
- As all the services are placed inside the kernel, they have a single address space
- It is bigger in size
- It is easy to implement/code
- Performance is high (As kernel can invoke any function directly as everything is placed in the kernel)
- Less Secure (If one service fails, entire system crashes)

Monolithic Architecture

Figure 3 Monolithic operating system kernel architecture.

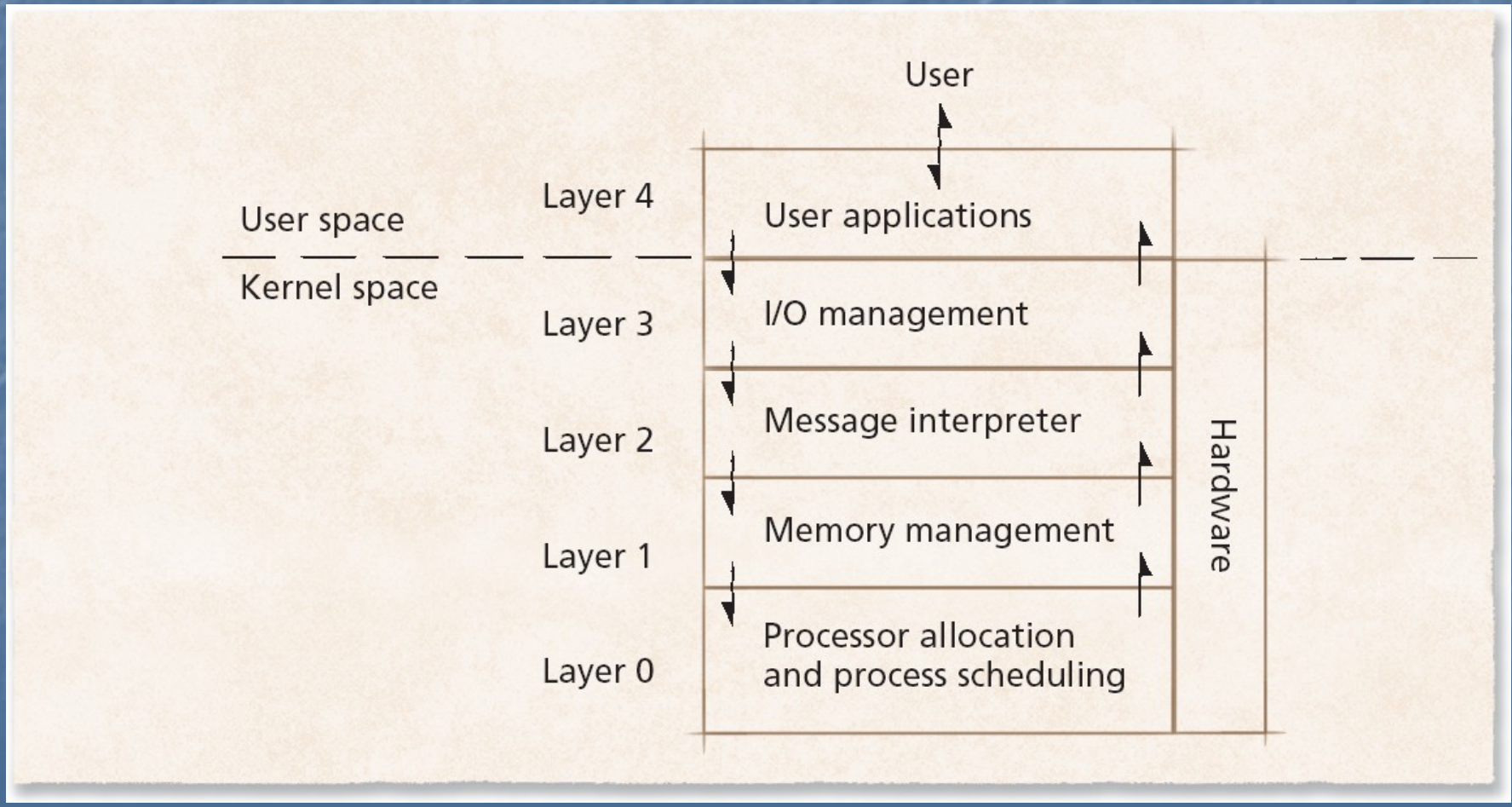


Layered Architecture

- Layered approach to operating systems
  - Tries to improve on monolithic kernel designs
    - Groups components that perform similar functions into layers
  - Each layer communicates only with layers immediately above and below it
  - Processes' requests might pass through many layers before completion
  - System throughput can be less than monolithic kernels

Layered Architecture

Figure 4 Layers of the operating system.



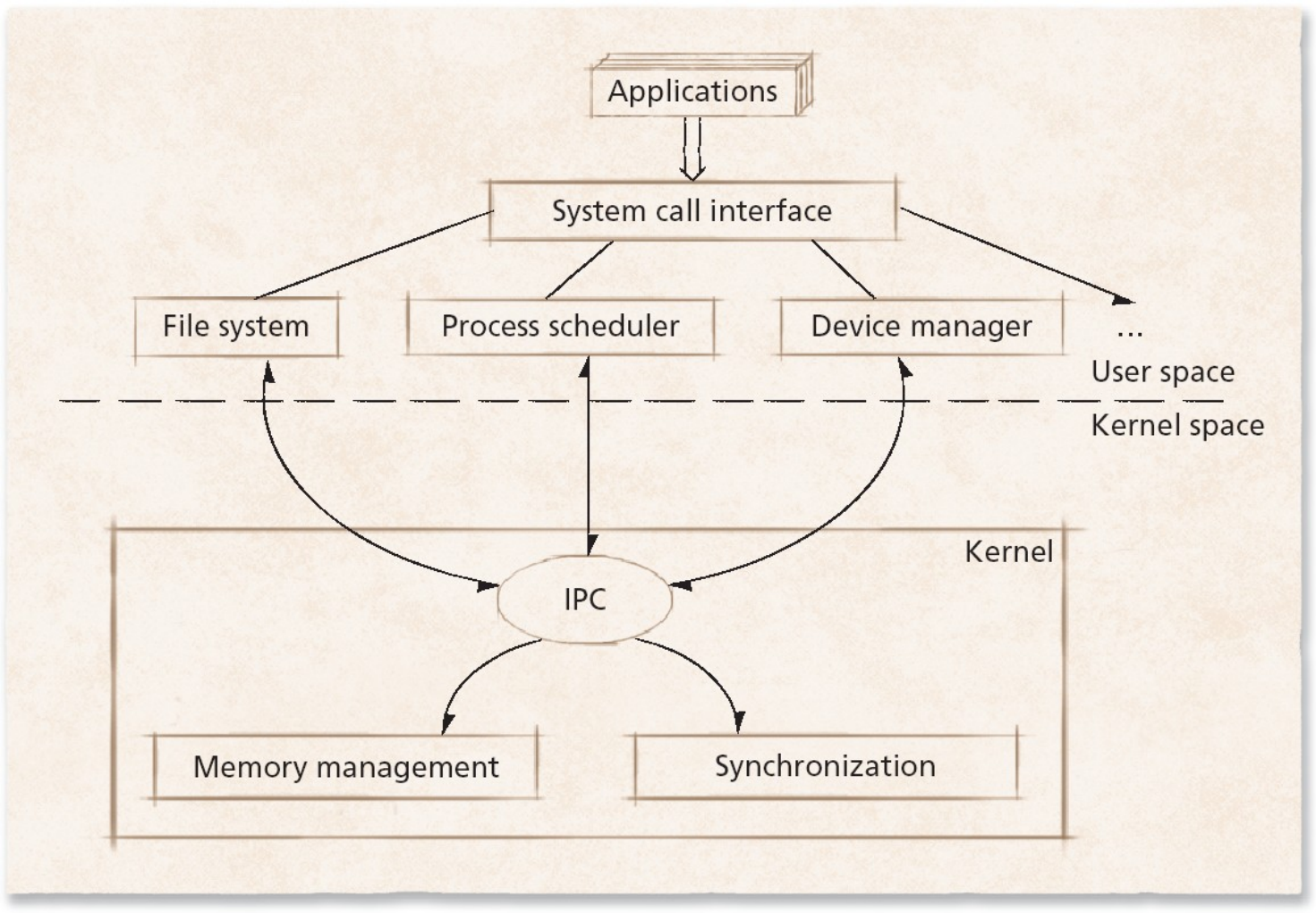
Microkernel Architecture

- Microkernel operating system architecture
  - Provides only small number of services
    - Attempt to keep kernel small and scalable

- Only bare minimum code is placed inside the kernel (only basic memory management and Inter Process Communication code)
- Here the kernel is broken down into processes called as servers
- As services(Servers provide services) are separated they have different address spaces
- It is smaller in size
- It is tough to implement/code
- Performance is low (As servers are separated, so to invoke services from other servers IPC(Inter Process Communication) is needed which requires kernel's permission and thus increases access time and lowers the performance)
- More Secure (Even if one service crashes, others can function properly because of separation)

Microkernel Architecture

Figure 5 Microkernel operating system architecture.



Basic	In microkernel user services and kernel, services are kept in separate address space.	In monolithic kernel, both user services and kernel services are kept in the same address space.
Size	Microkernel are smaller in size.	Monolithic kernel is larger than microkernel.
Execution	Slow execution.	Fast execution.
Extendible	The microkernel is easily extendible.	The monolithic kernel is hard to extend.
Security	If a service crashes, it does effect on working of microkernel.	If a service crashes, the whole system crashes in monolithic kernel.
Code	To write a microkernel, more code is required.	To write a monolithic kernel, less code is required.
Example	QNX, Symbian, L4Linux, Singularity, K42, Mac OS	Linux, BSDs (FreeBSD, OpenBSD, NetBSD), Microsoft Windows

Monolithic Kernel based Operating System

Microkernel based Operating System

